

# Measuring Muscle, Automatically

Geometry-aware deep learning for muscle architecture  
estimation in B-mode ultrasound

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CSE 754 Deep Learning for Computer Vision

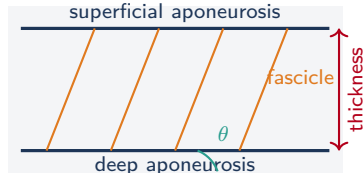
# The problem

Three numbers describe muscle architecture:

- **muscle thickness** (mm)
- **pennation angle** (degrees)
- **fascicle length** (mm)

They are still measured **by hand**: slow, subjective, and dependent on operator experience.

**Goal.** Produce all three automatically from a single ultrasound frame.



# What the task actually is

The data does not make this a supervised regression problem.

## No measurement labels

Training data gives *masks* only, never a thickness or an angle. The three targets must be **reconstructed geometrically**.

## No scale metadata

Two targets are in millimetres; the network sees pixels. The conversion must be **read off the scanner's own depth ruler**.

## Silent data defects

- 70% of frames disagree in size with their own mask
- 26% of fascicle images are duplicates

Neither raises an error. Both corrupt results in ways that look like success.

A naive split leaks **179** duplicate groups into validation. Content-grouped splitting reduces that to **0**.

# Pipeline



## Data

1 048 aponeurosis frames  
2 761 fascicle frames  
309 unlabelled test frames

## Model

U-Net, ResNet-34 encoder  
512×512, BCE + Dice  
94 min on one GPU

## Contribution

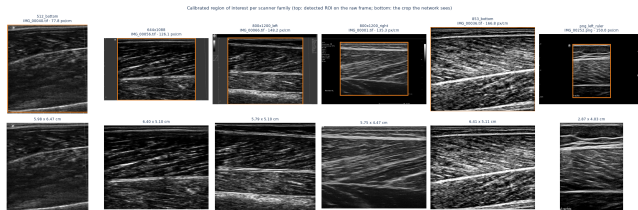
scale recovery  
dense orientation head  
geometry in mm space

# Scale calibration

Six scanner families, each drawing its depth ruler differently: left edge, right edge, bottom edge; ticks every 5 or 10 mm.

**100%** of the 309 test frames calibrate

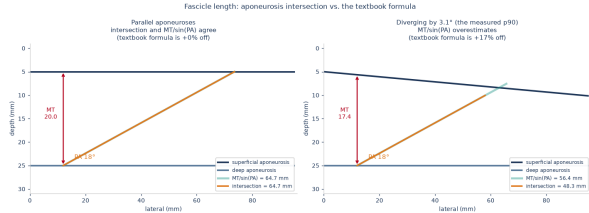
Validated independently: implied field of view spans **2.97** to **6.95** cm deep, matching the documented 3 to 7 cm scanner presets. Nothing was fitted to those numbers.



Detected region of interest per scanner family, and the crop the network sees.

# Geometric reconstruction

- Aponeuroses fitted as **curves**, not lines
- Pennation referenced to the **deep aponeurosis**, as the clinical definition specifies
- Orientation predicted as a dense  **$(\cos 2\theta, \sin 2\theta)$  field**, so that averaging directions is well defined



Two ways to reconstruct fascicle length. Which is better turned out to be the project's most interesting question.

Orientation targets are *derived*, so they were checked against an independent estimator: median disagreement **1.1** degrees, 100% of frames within 5 degrees.

# Evaluating without any measurement labels

No frame carries an expert measurement, so error against a human cannot be computed anywhere.

**Pseudo-ground-truth protocol.** Run the *same* geometry module twice, once on the annotator's masks and once on the network's, and compare.

## What it measures

the error the segmentation contributes, with the geometry held fixed.

## What it does not measure

error of the geometry module itself against a human. That needs expert annotations this data does not contain.

**External check.** Pennation angle is scale-invariant, so the reconstruction run on annotator masks can be compared with the outside world without trusting the calibration: it gives a median of **17.0** degrees, against a published range of 10 to 25 degrees and **16.4** degrees for an independently built pipeline.

## Segmentation, held out

|             | aponeurosis  | fascicle     |
|-------------|--------------|--------------|
| Dice        | <b>0.839</b> | 0.274        |
| IoU         | 0.731        | 0.161        |
| Orientation |              | <b>2.74°</b> |

The fascicle Dice is low because annotators traced a *representative sample* of fascicles, not all of them. A model finding a different valid set is penalised while being correct for the purpose.

## Geometry, predicted vs annotator masks

|                  |                      |
|------------------|----------------------|
| Muscle thickness | <b>0.3% relative</b> |
| Pennation angle  | <b>1.16°</b>         |
| Fascicle length  | 6.0% relative        |

Thickness is recovered from *predicted* masks to within 0.3% of what the annotator's own masks give. That single fact drove the project's main design decision.

Caveat: all 78 paired frames come from one scanner family.

| Proposed                         | Measured against what it replaced                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------|
| Dense orientation head           | Does not beat it. $0.79^\circ$ against <b><math>0.72^\circ</math></b> for the classical structure tensor. |
| Sequence smoothing               | Nearly inert. Within-run spread was already 0.02 mm in thickness.                                         |
| Per-frame confidence             | Does not predict error. Spearman $-0.08$ to $-0.19$ , not significant.                                    |
| Intersection for fascicle length | Helps, but not for the proposed reason. See next slide.                                                   |

The confidence measure initially *appeared* to detect failures. It was being told about them: an earlier version docked confidence whenever an estimate hit a clamp, so any correlation with error was circular. The term was removed.

## The main finding

**A model that is more faithful in principle can be a worse estimator in practice, when it depends on noisier inputs.**

The proposal argued that  $FL = MT/\sin(PA)$  is wrong because the aponeuroses are not parallel, and that intersecting the fascicle with the fitted curves is better. The premise is correct. The conclusion does not follow.

| Construction           | Median error |
|------------------------|--------------|
| Intersection           | 0.077        |
| Textbook $MT/\sin(PA)$ | <b>0.060</b> |

Wilcoxon  $p = 0.0003$ ,  $n = 78$ .

**Why.** Not because divergence is estimated badly: it is recovered to  $0.15^\circ$ . The textbook formula leans on *thickness*, recovered to 0.3%. The intersection leans on a *difference of two fitted angles*, inheriting angular noise twice over.

What the intersection actually contributes is **boundedness**: it cannot diverge at shallow pennation, where  $MT/\sin(PA)$  runs away as  $1/\sin$ . Right mechanism, wrong reason.

## What worked

- Thickness to **0.3%**, pennation to **1.16°**
- Aponeurosis Dice **0.839**
- Scale recovered on **100%** of test frames

The component that mattered most was **scale recovery**, which the original proposal never contemplated, because it assumed the measurements were learnable targets.

## Honest limitations

- No expert-measurement validation anywhere
- Paired set is 78 frames from one scanner family
- Fascicle length is extrapolated, not observed
- Single split, single seed

Code, report and full ablations: [github.com/silvererudite/umud-muscle-architecture](https://github.com/silvererudite/umud-muscle-architecture)